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Maximum Subarray Sum

[2, 3, -8, 7, -1, 2, 3]

Sum = 0 2 5 ~~-3~~ 0 7 6 8 11

maxSum = ~~-∞~~ 2 5 7 8 11

Steps

1) add ele to sum
2) maxSum update
3) if(sum<0) sum=0

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arr = { -2, -4, -1, -7, -6 }

sum = 0 -2 0 -4 0 -1 0 -7 0 -6 0

maxSum = -0 -2 -1

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Maximum Product Subarray

[-2, 6, -3, -10, 0, 2]

product = 1 -2 -12 36 -360 1 2

maxPro = -10 -2 36 $\boxed{180}$

pro = 1 2 1 -10 30 180 -360

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$\boxed{-2 \quad 6 \quad -3 \quad -10}$ 0 $\boxed{2 \quad 1 \quad 3}$

entire pro is -ve

$\boxed{\times \quad \times \times \times}$ 0 $\boxed{\times \times \times}$ 0 $\boxed{\times \times \times}$

even no. of -ve = +ve

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arr = {-1}

pro = 1 -1

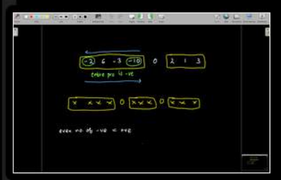
max = ~~-∞~~ -1

-1, 0, -2, 0, -3

arr = {-1, 0, -2}

pro = 1 -1 1 -2

max = -∞ -1 0



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Majority Element

Moore's Voting Algorithm

1 2 2 1 3 1 1 1 4 1

winner = ~~1~~ ~~2~~ ~~3~~ 1

count = ~~1~~ ~~0~~ 1 ~~0~~ 1 ~~0~~ ~~1~~ ~~2~~ ~~1~~ 2

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arr = 2 3 2 3 2 3 2 3 1 1 1 1

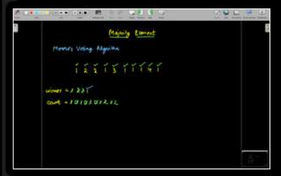
winner = 2 1 α

count = 1 0 1 0 1 0 1 0 1 2 3 4

Reason :

10 elements (>6)

1st half	2nd half
2 ones	4 ones
1 ones	5 ones



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Sort 0's 1's and 2's

(Dutch Flag Algorithm)

				l				h			
0	0	0	0	1	1	1	1	1	2	2	2
									m		

check arr[mid] = 0, 1, 2

- 0 → swap with lo; lo++ mid++
- 1 → mid++
- 2 → swap with hi; hi--

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lo, mid, hi

lo m hi

0.. 0 0 0 0 0 1 1 1 1..1 x x x x x x x x x 2 2 2 2 2 2 . . .2

0, 1, 2 unsorted

0 to lo-1 → 0

lo to mid-1 → 1

hi+1 to n-1 → 2